

## CS 464 ARTIFICIAL INTELLIGENCE

### MODEL QUESTION BANK

#### MODULE 1

1. Discuss the history of Artificial Intelligence.
2. Why knowledge representation is necessary in AI systems? Give AI systems in which knowledge is important?
3. Define: (a) intelligence, (b) artificial intelligence, (c) agent, (d) rationality, (e) logical reasoning.
4. There are well-known classes of problems that are intractably difficult for computers, and other classes that are provably undecidable. Does this mean that AI is impossible?
5. Explain the relationship between evolution and one or more of autonomy, intelligence, and learning.
6. Explain the concept of rationality.
7. Let us examine the rationality of various vacuum-cleaner agent functions.
  1. Show that the simple vacuum-cleaner agent function described in Figure 2.3 is indeed rational under the assumptions listed on page
  2. Describe a rational agent function for the case in which each movement costs one point. Does the corresponding agent program require internal state?
  3. Discuss possible agent designs for the cases in which clean squares can become dirty and the geography of the environment is unknown. Does it make sense for the agent to learn from its experience in these cases? If so, what should it learn? If not, why not?
8. What are the properties of task environments? Explain.
9. What do you understand by structure of an Agent?
10. What is agent program? Explain four basic kinds of agent programs.

#### MODULE 2

1. Define state, state space, search tree, search node, goal, action, successor function and branching factor.
2. Explain why problem formulation must follow goal formulation.
3. What is a well defined problem?
4. Give a complete problem formulation for each of the following problems. Choose a formulation that is precise enough to be implemented.
  1. There are six glass boxes in a row, each with a lock. Each of the first five boxes holds a key unlocking the next box in line; the last box holds a banana. You have the key to the first box, and you want the banana.
  2. You start with the sequence ABABAECCCEC, or in general any sequence made from A, B, C, and E. You can transform this sequence using the following equalities:  $AC = E$ ,  $AB = BC$ ,  $BB = E$ , and  $Exx = xx$  for any  $xx$ . For example, ABBC can be transformed into AEC, and then AC, and then E. Your goal is to produce the sequence E.
  3. There is an  $n \times n \times n$  grid of squares, each square initially being either unpainted floor or a bottomless pit. You start standing on an unpainted floor square, and can either paint the square under you or move onto an adjacent unpainted floor square. You want the whole floor painted.
  4. A container ship is in port, loaded high with containers. There 13 rows of containers, each 13 containers wide and 5 containers tall. You control a crane that can

- move to any location above the ship, pick up the container under it, and move it onto the dock. You want the ship unloaded.
5. Show that the 8-puzzle states are divided into two disjoint sets, such that any state is reachable from any other state in the same set, while no state is reachable from any state in the other set.
  6. Explain uninformed search and informed search with example
  7. Compare informed and uninformed search.
  8. Explain uninformed BFS, DFS, and Iterative deepening DFS
  9. Explain heuristic function.
  10. Illustrate the problem of under estimation and over estimation in A\*
  11. Explain heuristic search technique with example.
  12. Determine whether goal driven or data driven search would be preferable for solving error in a computer. State the reason also.
  13. Differentiate between A\* algorithm and Best First Search algorithm.
  14. Prove each of the following statements:
    - a. BFS is a special case of uniform-cost search.
    - b. BFS, DFS, and uniform-cost search are special cases of best-first search.
    - c. Uniform-cost search is a special kind of A\* search.
  15. Distinguish between data driven search and goal driven search strategies.
  16. Define uninformed search. Which kind of problems can use uninformed search?  
Why it can't be used in all AI problems?
  17. What is A\* algorithm? Is it optimal under all conditions?
  18. How is AO\* different from A\* algorithm?

### MODULE 3

1. Explain the minimax algorithm.
2. How Minimax procedure is implemented in exhaustively searchable state spaces?  
Explain using any two person game.
3. How and when heuristic is used in Minimax search technique? Illustrate the usage of heuristic in Minimax procedure
4. What is the importance of two bounds in Alpha-Beta cut-offs.
5. Explain Alpha-beta search algorithm.
6. Define the terms constraint satisfaction problem, constraint, backtracking search, arc consistency, backjumping and min-conflicts
7. Explain why it is a good heuristic to choose the variable that is most constrained, but the value that is least constraining in a CSP search.
8. Solve the following crypt arithmetic problem stating all the constraints. EAT + THAT = APPLE
9. Using Constraint Satisfaction algorithm solve the following Crypt Arithmetic problem  
 SEND  
 + MORE  
 -----  
 MONEY
10. Explain back tracking in CSP with an example.
11. Give a precise formulation of the following constraint satisfaction problem:

Class scheduling: There is a fixed number of professors and classrooms, a list of classes to be offered, and a list of possible time slots for classes. Each professor has a set of classes that he or she can teach.

## MODULE 4

1. What is logical agent? How is it different from knowledge base d agents?
2. Explain the BNF grammar of sentences in propositional logic.
3. Explain the syntax and semantics of first order logic.
4. What is an atomic sentence?
5. Consider a knowledge base containing just two sentences:  $P(a)P(a)$  and  $P(b)P(b)$ . Does this knowledge base entail  $\forall x P(x)\forall x P(x)$ ? Explain your answer in terms of models.
6. Is the sentence  $\exists x, y \ x y$  valid? Explain.
7. Is the sentence  $\exists x, y \ x=y$  valid? Explain.
8. Does the fact  $\neg \text{Spouse}(\text{George}, \text{Laura}) \neg \text{Spouse}(\text{George}, \text{Laura})$  follow from the facts  $\text{Jim} \neq \text{George} \text{Jim} \neq \text{George}$  and  $\text{Spouse}(\text{Jim}, \text{Laura}) \text{Spouse}(\text{Jim}, \text{Laura})$ ? If so, give a proof; if not, supply additional axioms as needed. What happens if we use  $\text{SpouseSpouse}$  as a unary function symbol instead of a binary predicate?
9. Translate into first-order logic the following sentences:
  1. Understanding leads to friendship.
  2. Friendship is transitive.Remember to define all predicates, functions, and constants you use.
10. What are the inference rules for quantifiers? Explain.
11. What is unification and lifting?
12. Explain forward chaining with an example.
13. Explain backward chaining with example.
14. Convert the following sentences into first order logic:

Everyone who loves all animals is loved by someone.  
Anyone who kills an animal is loved by no one.  
Jack loves all animals.  
Either Jack or Curiosity killed the cat, who is named Tuna.  
Did Curiosity kill the cat?

Give a resolution proof to answer the question “Did Curiosity kill the cat?”
15. Explain the BNF grammar of sentences in propositional logic.
16. Explain the syntax and semantics of first order logic.

## MODULE 5

1. What are the designing issues of a learning element?
2. Differentiate supervised, unsupervised and reinforcement learning.
3. Explain inductive learning.
4. We never test the same attribute twice along one path in a decision tree. Why not?
5. Explain expressiveness of decision trees.
6. How do we choose attributes for decision tree learning?
7. Define information gain.
8. Explain decision tree algorithm. Consider the following data set comprised of two binary input attributes (A1 and A2) and one binary output.

| Example | A1 | A2 | Output Y |
|---------|----|----|----------|
| x1      | 1  | 1  | 1        |
| x2      | 1  | 1  | 1        |
| x3      | 1  | 0  | 0        |
| x4      | 0  | 0  | 1        |
| x5      | 0  | 1  | 0        |
| x6      | 0  | 1  | 0        |

Use the DECISION-TREE-LEARNING algorithm to learn a decision tree for these data. Show the computations made to determine the attribute to split at each node.

9. Consider the following data set comprised of three binary input attributes (A1 , A2 , and A3) and one binary output.

| Example        | A <sub>1</sub> | A <sub>2</sub> | A <sub>3</sub> | Output y |
|----------------|----------------|----------------|----------------|----------|
| x <sub>1</sub> | 1              | 0              | 0              | 0        |
| x <sub>2</sub> | 1              | 0              | 1              | 0        |
| x <sub>3</sub> | 0              | 1              | 0              | 0        |
| x <sub>4</sub> | 1              | 1              | 1              | 1        |
| x <sub>5</sub> | 1              | 1              | 0              | 1        |

Use the DECISION-TREE-LEARNING algorithm to learn a decision tree for these data. Show the computations made to determine the attribute to split at each node.

10. What is multivariate linear regression? Explain.
11. Describe the various forms of learning?
12. State and explain Ockham's razor principle
13. Explain Linear classification with logistic regression